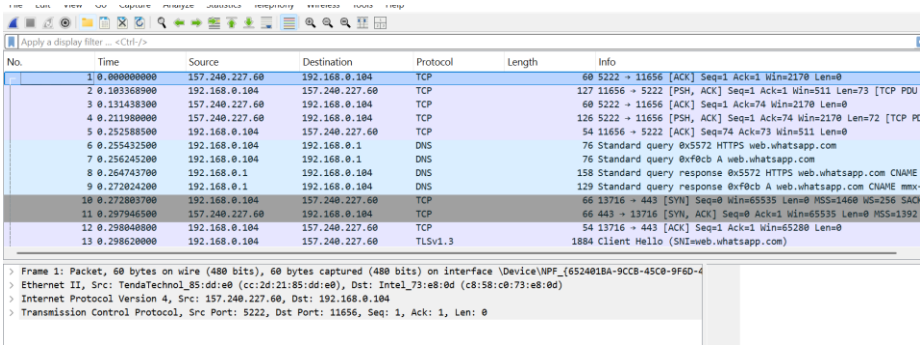
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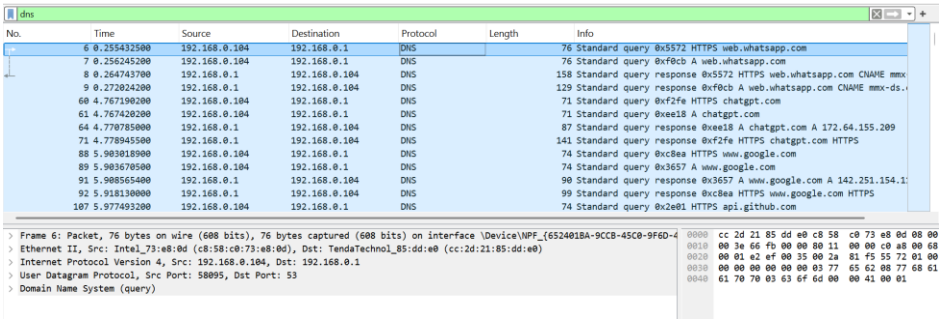
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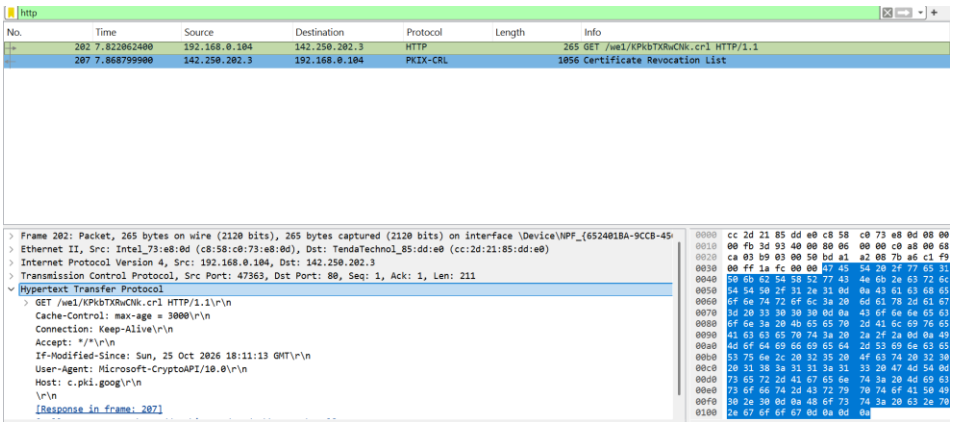
Cyber Security Internship Report

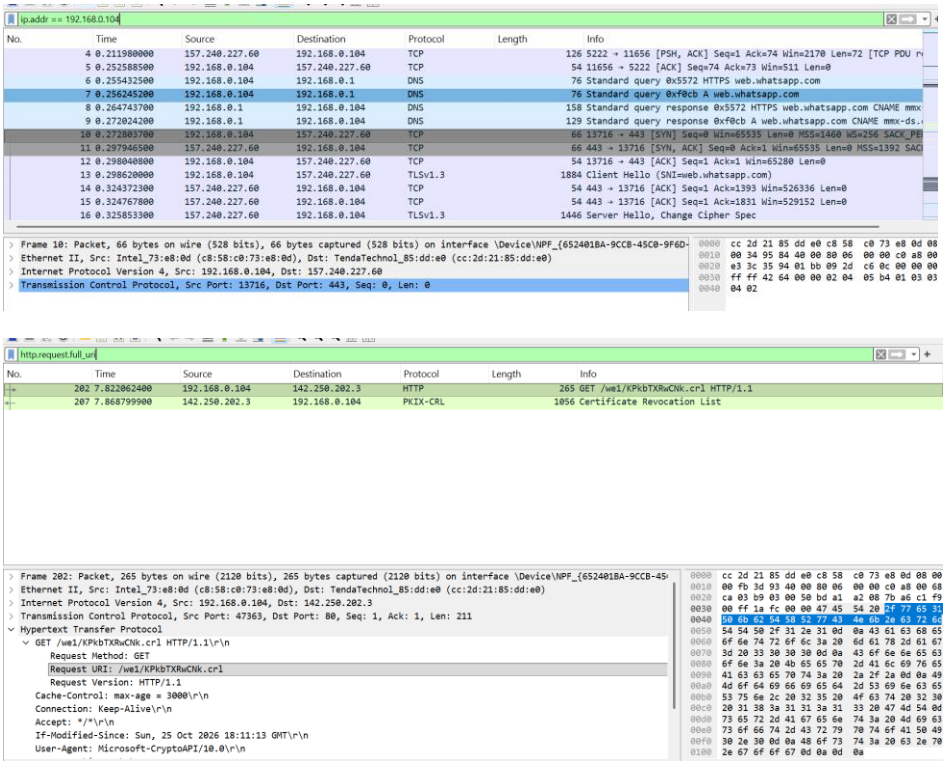
WEEK 03 TASK 09

Network Traffic Analysis — Wireshark Basics









1. Objective

The objective of this task was to learn how to capture and analyze network traffic using Wireshark. The practical focused on understanding network packets, identifying protocols, applying display filters, inspecting IP addresses, and recognizing suspicious or insecure network communication.

2. Introduction

Wireshark is a network protocol analyzer used to capture and examine network traffic. It helps security analysts understand communication between devices, identify protocols, investigate network issues, and detect potentially suspicious activities.

3. Practical Implementation

3.1 Network Traffic Capture

Wireshark was opened and a network interface was selected to capture live network traffic. A short capture was performed while browsing a website.

The captured packets displayed information such as packet number, time, source IP, destination IP, protocol, length, and packet details.

3.2 DNS Display Filter

The following display filter was used:

dns

This filter displays DNS packets and helps identify domain name queries and responses. During the capture, DNS queries for web.whatsapp.com were observed.

3.3 HTTP Display Filter

The following filter was tested:

http

This filter is used to display HTTP communication. Modern websites commonly use HTTPS, so HTTP request packets were not successfully verified in the available capture.

3.4 IP Address Display Filter

The following filter was successfully applied:

ip.addr == 192.168.0.104

This filter displays packets where the specified IP address appears as either the source or destination.

The following network addresses were identified:

- Local Device IP: 192.168.0.104
- DNS Server / Router: 192.168.0.1
- Remote Server IP: 157.240.227.60

4. Packet Analysis and Observations

The captured network traffic contained TCP and DNS packets.

The following observations were made:

- DNS queries were sent for web.whatsapp.com.
- DNS responses were received from the local DNS server.

- TCP communication was observed between the local device and a remote server.
- Communication involving port 443 was observed, indicating HTTPS-related encrypted traffic.

An HTTP request packet was not successfully verified during this practical because the available traffic primarily showed DNS and TCP communication.

5. Suspicious Traffic and Plaintext Password Analysis

Plaintext communication is a security risk because sensitive information transmitted without encryption may be readable by someone who can access the network traffic.

For example, an insecure HTTP login request could contain:

POST /login HTTP/1.1

username=demo&password=Test123

This is an illustrative example, not an actual captured password.

Potential indicators of suspicious network activity include:

- Passwords transmitted through unencrypted HTTP.
- Repeated failed connection attempts.
- Unexpected connections to unknown IP addresses.
- Unusual DNS queries.
- Unexpected data transfers.

Security Recommendations:

- Use HTTPS and TLS encryption.
- Enable Multi-Factor Authentication (MFA).
- Avoid transmitting sensitive information over unencrypted connections.
- Monitor network traffic regularly.
- Investigate unusual network connections.

6. Learning Outcomes

Through this practical exercise, I learned:

- How to open Wireshark and capture network traffic.
- How to identify source and destination IP addresses.
- How to apply DNS, HTTP, and IP address display filters.
- How to recognize DNS and TCP communication.
- How encrypted HTTPS traffic differs from unencrypted HTTP traffic.
- Why plaintext passwords create security risks.
- How Wireshark supports network monitoring and security investigation.

7. Limitations

The available capture primarily contained DNS and TCP traffic. An HTTP request packet and actual plaintext credentials were not successfully captured or verified. Therefore, plaintext password analysis was covered through a security example rather than an actual captured credential.

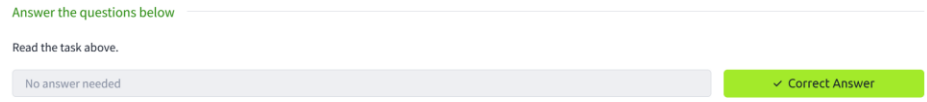
8. Conclusion

This practical provided hands-on experience with Wireshark and basic network traffic analysis. I successfully captured network packets, identified DNS and TCP communication, and applied an IP address filter. I also learned about HTTP, HTTPS encryption, plaintext password risks, and suspicious network activity. Wireshark is an important tool for network troubleshooting, monitoring, and SOC security investigations.

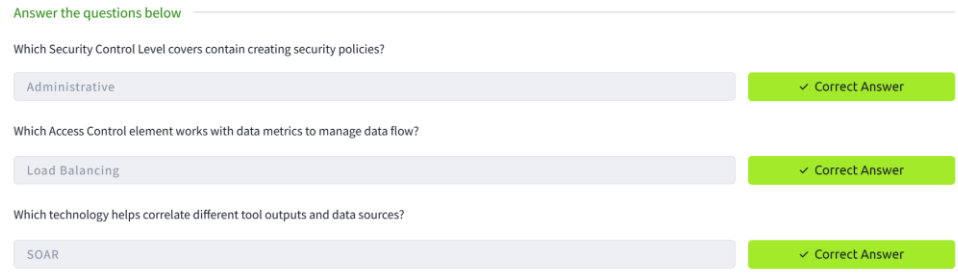
TRY HACK ME



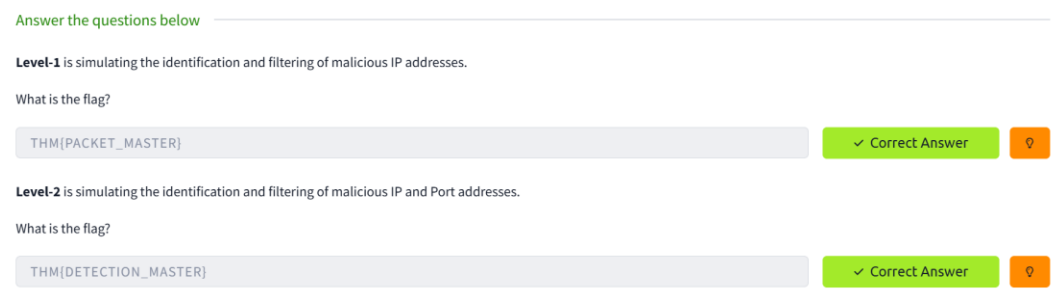
T1



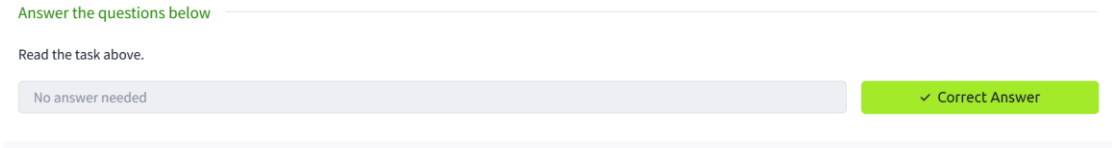
T2



T3



T4





Task 1: Introduction

Network security is the practice of protecting networks, connected devices, and data from unauthorized access, misuse, and cyber threats. Network traffic analysis helps security professionals monitor communication between devices, identify unusual activities, and investigate potential attacks.

This room introduced the basic concepts of network security and traffic analysis, including identifying suspicious IP addresses and ports.

Task 2: Network Security and Network Data

Network security uses different technologies and security controls to protect information and maintain safe communication.

Key concepts covered:

- Administrative Controls: Security policies, procedures, and guidelines that help organizations manage security.
- Load Balancing: Distributes network traffic across multiple systems to improve performance and availability.
- SOAR (Security Orchestration, Automation and Response): Helps security teams coordinate tools, automate workflows, and respond to security incidents.
- Network Data: Information transferred between devices, including source and destination IP addresses, ports, protocols, and packet details.

Understanding network data helps security analysts recognize normal communication and investigate suspicious activity.

Task 3: Traffic Analysis

Traffic analysis involves monitoring, recording, and examining network communications to identify performance issues, unusual patterns, and possible security threats.

Two common approaches are:

- Flow Analysis: Examines traffic summaries and statistics, such as communication between IP addresses, traffic volume, and connection patterns.
- Packet Analysis: Examines individual packets in greater detail, including headers, protocols, and available packet contents. Deep Packet Inspection (DPI) can provide more detailed information.

Practical Activity

The TryHackMe simulation introduced the identification and filtering of suspicious IP addresses and network ports.

Level 1 – Malicious IP Address Identification

Flag: THM{PACKET_MASTER}

Level 2 – Malicious IP and Port Identification

Suspicious ports covered in the provided solution:

- Port 2222
- Port 7777
- Port 4444

Flag: THM{DETECTION_MASTER}

The activity demonstrated how filtering suspicious network indicators can help narrow down potentially malicious traffic for further investigation.

Task 4: Conclusion

This room improved my understanding of network security, network data, and traffic analysis. I learned the difference between flow analysis and packet analysis and understood how suspicious IP addresses and ports can be investigated through filtering.

I also learned that tools and technologies such as Wireshark, IDS/IPS, SIEM, and SOAR support network monitoring, threat detection, and incident response. Traffic analysis is an important part of cybersecurity because it helps identify unusual communication and investigate potential network attacks.